

Credit Card Fraud Detection

1. Dataset:

The dataset used in this project is the **Credit Card Fraud Detection dataset from Kaggle**. It contains transactions made by European cardholders over a period of two days.

- The dataset consists of **284,807 transactions (rows)** and **30 features (columns)**.
- Features include:
 - **Time** and **Amount**
 - **V1 to V28** (anonymized features obtained using PCA for privacy)
 - **Class** (target variable)
 - 0 → Normal transaction
 - 1 → Fraudulent transaction

◆ Imbalanced Dataset

The dataset is highly **imbalanced**:

- Normal transactions ≈ **99.8%**
- Fraud transactions ≈ **0.2%**

This imbalance makes fraud detection challenging, as the model must correctly identify rare fraud cases without misclassifying normal transactions

2. Methodology:

The project follows a structured machine learning pipeline:

◆ **Data Preprocessing**

- Loaded dataset using Pandas
- Separated features (X) and target (y)
- Checked for missing values and ensured data consistency

◆ **Scaling**

- Applied **Standard Scaler** to normalize the data
- Scaling is important because:
 - Features like *Amount* have large values
 - Models perform better with normalized inputs

◆ **Train-Test Split**

- Data was divided into:
 - **80% Training data**
 - **20% Testing data**
- This ensures proper evaluation of the model on unseen data

3. Model Used:

The primary model used in this project is:

◆ **Logistic Regression**

- A supervised machine learning algorithm used for **binary classification**
- It predicts the probability of a transaction being fraud or normal

◆ **Why Logistic Regression?**

- Simple and efficient
- Works well with large datasets
- Provides probability outputs
- Suitable for baseline fraud detection

Overview of Logistic Regression:

Logistic Regression is a **supervised machine learning algorithm** used for **binary classification problems**, where the output is one of two classes.

In this project:

- **Class 0 → Normal Transaction**
- **Class 1 → Fraudulent Transaction**

The model predicts the **probability** of a transaction being fraudulent and classifies it based on a threshold value.

4. Results:

The model performance was evaluated using multiple metrics:

◆ **Confusion Matrix**

- Shows correct and incorrect predictions:
 - True Positives (Fraud correctly detected)
 - True Negatives (Normal correctly detected)
 - False Positives and False Negatives

◆ **Precision**

- Measures how many predicted frauds are actually fraud
- High precision → fewer false alarms

◆ **Recall**

- Measures how many actual frauds are detected
- Important in fraud detection (don't miss fraud cases)

◆ **F1-Score**

- Harmonic mean of precision and recall
- Provides balanced performance evaluation

👉 The model achieved **high accuracy and good fraud detection capability**, although recall can be improved due to dataset imbalance.

5. Deployment

The model was deployed using **Streamlit**, a Python framework for building interactive web applications.

◆ **Features of the Web App:**

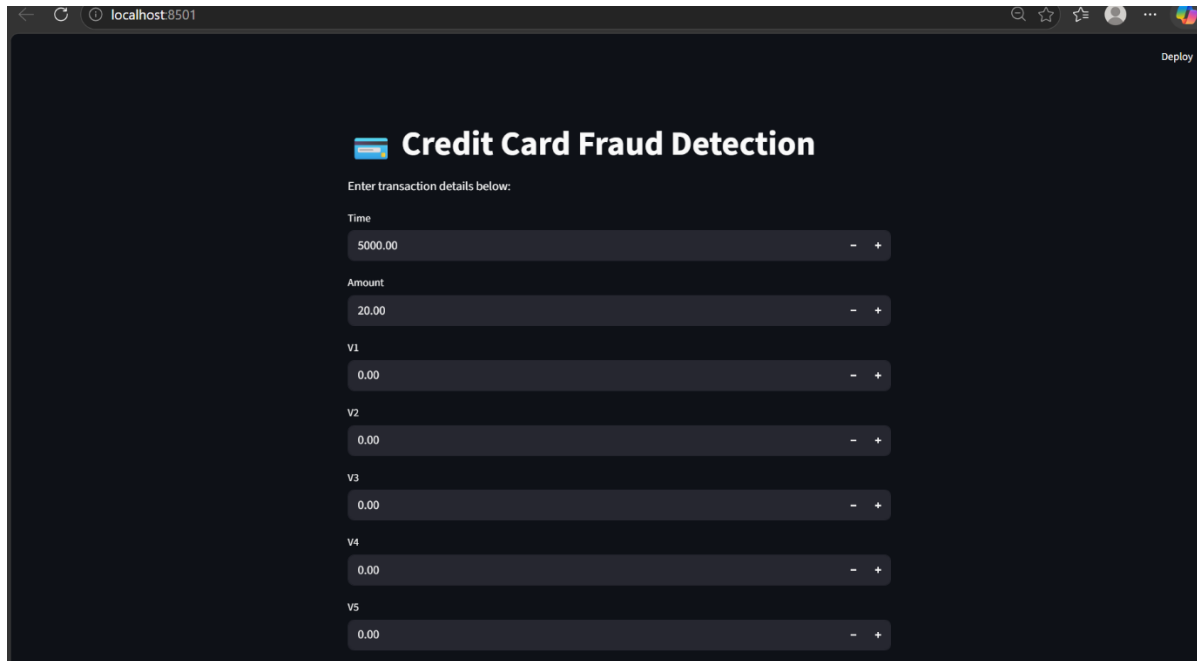
- User can input transaction details (Time, Amount, V1–V28)
- Model processes input in real time
- Displays prediction:
 -  Normal Transaction
 -  Fraudulent Transaction

◆ **Advantages:**

- Easy to use interface
- Real-time fraud detection
- Demonstrates practical application of machine learning

6.Outputs:

Normal Prediction:



The screenshot shows a web browser window at localhost:8501. The page has a dark theme and a title bar with standard browser controls. The main heading is 'Credit Card Fraud Detection' with a credit card icon. Below it, a prompt says 'Enter transaction details below:'. There are six input fields, each with a label and a 'Predict' button. The labels are Time, Amount, V1, V2, V3, and V4. The values entered are 5000.00, 20.00, 0.00, 0.00, 0.00, and 0.00 respectively. A 'Deploy' button is in the top right corner.

localhost:8501

Deploy

Credit Card Fraud Detection

Enter transaction details below:

Time
5000.00 - +

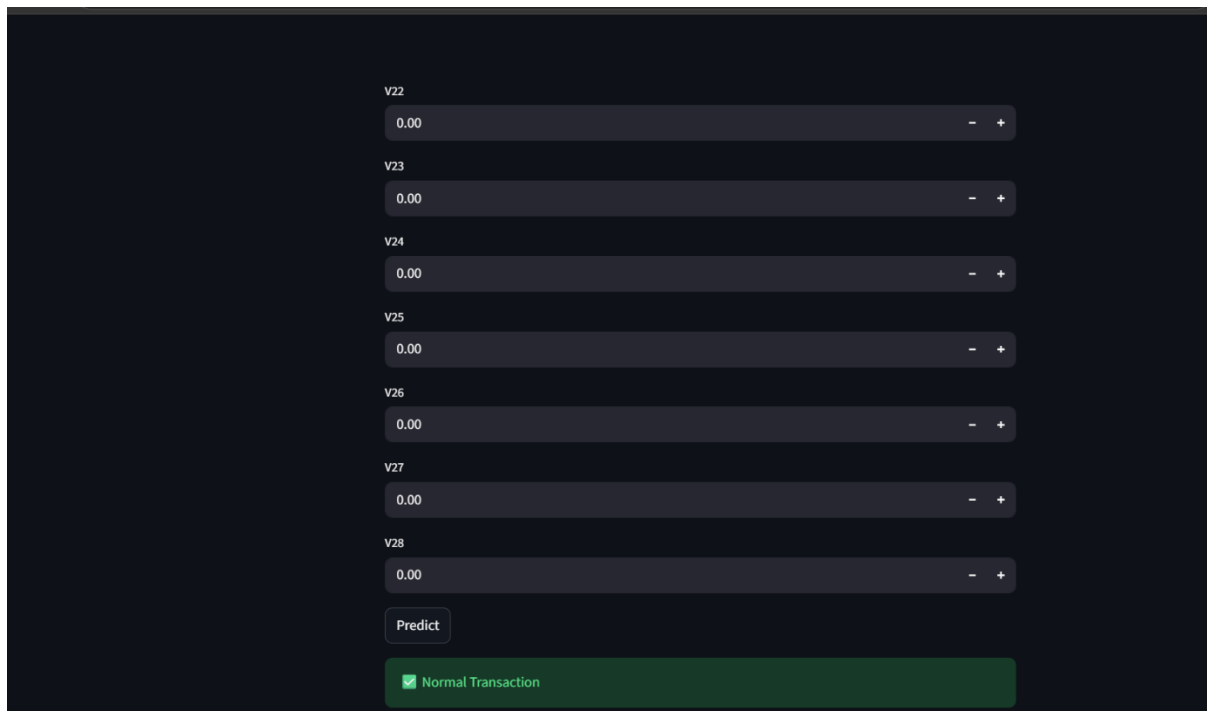
Amount
20.00 - +

V1
0.00 - +

V2
0.00 - +

V3
0.00 - +

V4
0.00 - +



This screenshot shows the same web application after a prediction. The input fields for V22 through V28 are visible, all with values of 0.00. A 'Predict' button is located below these fields. At the bottom, a green box displays the result: 'Normal Transaction' with a green checkmark icon.

V22
0.00 - +

V23
0.00 - +

V24
0.00 - +

V25
0.00 - +

V26
0.00 - +

V27
0.00 - +

V28
0.00 - +

Predict

✓ Normal Transaction

Fraud Prediction:

localhost8501

Deploy

 Credit Card Fraud Detection

Enter transaction details below:

Time

50000.00

-

+

Amount

2000.00

-

+

V1

-5.00

-

+

V2

3.00

-

+

V3

-4.00

-

+

V4

0.00

-

+

V5

0.00

-

+

V6

0.00

-

+

0.00

-

+

V22

0.00

-

+

V23

0.00

-

+

V24

0.00

-

+

V25

0.00

-

+

V26

0.00

-

+

V27

0.00

-

+

V28

0.00

-

+

Predict

⚠️ Fraudulent Transaction Detected!